

Exercise 7

AIM :

To write a C program for below requirements

1. function for SetBit(), ResetBit(), ReadBit(), ToggleBit(), by passing parameters and return parameters
2. Function to ReverseString(), IntToString() by passing array as parameter

ALGORITHM :

BIT MANIPULATION :

1. SetBit() ALGORITHM:

INPUT: num (8-bit integer), bit_position (0–7)

OUTPUT: Modified 8-bit integer with specified bit set to 1

STEPS:

1. Create a mask by left-shifting 1 by bit_position

mask = 1 << bit_position

2. Perform bitwise OR operation between num and mask

result = num | mask

3. Return result

2. ResetBit() ALGORITHM:

INPUT: num (8-bit integer), bit_position (0–7)

OUTPUT: Modified 8-bit integer with specified bit set to 0

STEPS:

1. Create a mask by left-shifting 1 by bit_position

mask = 1 << bit_position

2. Invert the mask using bitwise NOT

inverted_mask = ~mask

3. Perform bitwise AND operation between num and inverted_mask

`result = num & inverted_mask`

4. Return result

3. ReadBit() ALGORITHM:

INPUT: num (8-bit integer), bit_position (0–7)

OUTPUT: Value of specified bit (0 or 1)

STEPS:

1. Right-shift num by bit_position

`shifted_num = num >> bit_position`

2. Perform bitwise AND with 1

`bit_value = shifted_num & 1`

3. Return bit_value

4. ToggleBit() ALGORITHM:

INPUT: num (8-bit integer), bit_position (0–7)

OUTPUT: Modified 8-bit integer with specified bit flipped

STEPS:

1. Create a mask by left-shifting 1 by bit_position

`mask = 1 << bit_position`

2. Perform bitwise XOR operation between num and mask

`result = num ^ mask`

3. Return result

5. PrintBinary() ALGORITHM:

INPUT: num (8-bit integer)

OUTPUT: None (prints binary representation)

STEPS:

1. Print label "Binary: "
2. For i from 7 down to 0:
 - a. Call ReadBit(num, i)
 - b. Print the bit value
3. Print newline character

STRING MANIPULATION :

6. ReverseString() ALGORITHM:

INPUT: str[] (character array)

OUTPUT: None (modifies original string)

STEPS:

1. Find string length using strlen(str)
2. Set start = 0
3. Set end = length - 1
4. While start < end:
 - a. temp = str[start]
 - b. str[start] = str[end]
 - c. str[end] = temp
 - d. start++
 - e. end--

7. IntToString() ALGORITHM:

INPUT: num (integer), str[] (character array)

OUTPUT: None (stores converted string in str[])

STEPS:

1. Initialize i = 0, is_negative = 0
2. If num < 0:
 - a. is_negative = 1
 - b. num = -num

3. If num == 0:

a. str[i] = '0'

b. i++

4. Else:

While num > 0:

a. digit = num % 10

b. str[i] = digit + '0'

c. i++

d. num = num / 10

5. If is_negative == 1:

a. str[i] = '-'

b. i++

6. Set str[i] = '\0'

7. Call ReverseString(str)

MAIN FUNCTION :

STEPS:

1. Print header for bit manipulation section

2. Initialize num = 0b10101010 (170 decimal)

3. Print original number and its binary representation

4. Test SetBit():

a. Call SetBit(num, 0)

b. Print result and binary representation

5. Test ResetBit():

a. Call ResetBit(num, 7)

b. Print result and binary representation

6. Test ReadBit():

a. For bit positions 0 to 7:

- Call ReadBit(num, position)

- Print bit position and value

7. Test ToggleBit():

- Call ToggleBit(num, 3)
- Print result and binary representation

8. Print header for string manipulation section

9. Test ReverseString():

- str1 = "Hello World"
- Print original string
- Call ReverseString(str1)
- Print reversed string
- Call ReverseString(str1)
- Print restored string

10. Test IntToString():

- numbers[] = {123, -456, 0, 7890}
- Declare str_buffer[20]
- For each number:
 - Call IntToString(number, str_buffer)
 - Print integer
 - Print converted string
 - Call ReverseString(str_buffer)
 - Print reversed string

11. Return 0

OUTPUT :

```
Bit Manipulation Functions
Original number: 170
Binary: 10101010

After setting bit 0: 171
Binary: 10101011

After resetting bit 7: 42
Binary: 00101010

Reading bits from original number:
Bit 0: 0
Bit 1: 1
Bit 2: 0
Bit 3: 1
Bit 4: 0
Bit 5: 1
Bit 6: 0
Bit 7: 1

After toggling bit 3: 162
Binary: 10100010
```

String Manipulation Functions

Reversing the string

```
Original string: Hello World  
Reversed string: dlroW olleH  
Reversed back: Hello World
```

Integer to String Conversion

```
Integer: 123  
String: 123  
Reversed: 321
```

```
Integer: -456  
String: -456  
Reversed: 654-
```

```
Integer: 0  
String: 0  
Reversed: 0
```

```
Integer: 7890  
String: 7890  
Reversed: 0987
```